Run223 complete | Cosine32 to half
Each paired with a fresh unchanged control | seed 0 | solid unsmoothed curves | independent scales

MLP | Spark GB10 | 5,000 images
Cosine32 to half | base LR 0.001

ResNet-32 | RTX4090 | 50,000 images
Cosine32 to half | base LR 0.003

Transformer | L40S | 2,088,600 targets
Cosine32 to half | base LR 0.003

Training loss (log)
Raw $\rho_t$
$\|g_t\|$
$\eta\hat{\kappa}_t$
Applied/base LR
Unchanged
Cosine32 to half
Worker switch
(exact replay)
Completed optimizer updates

Gray region: shared prefix; vertical line: intervention fork. Transformer cosine switches L40S workers at544 after exact replay.
Other curves stay on one worker. Historical control reuse failed in all three cases; no historical bitwise replay is claimed.